

Project Manual

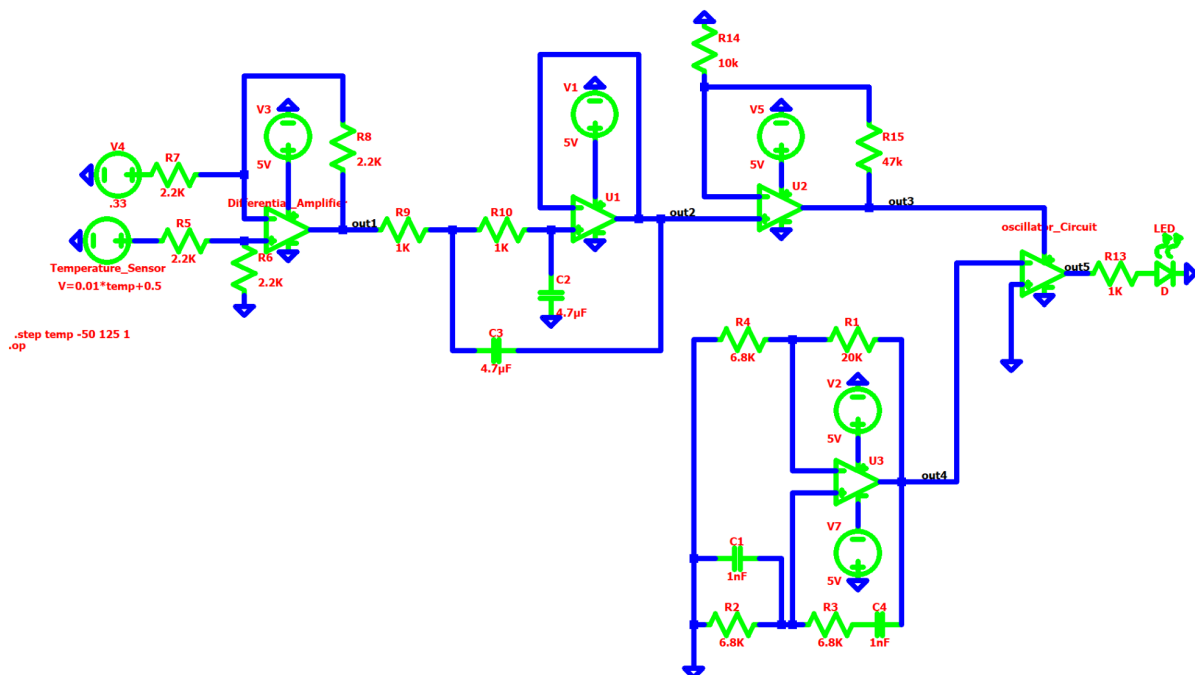
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Schematic Link:

https://drive.google.com/file/d/1KSVNnlbaQ1_E99u4sSHttALh9m57Ts3J/view?usp=drive_link

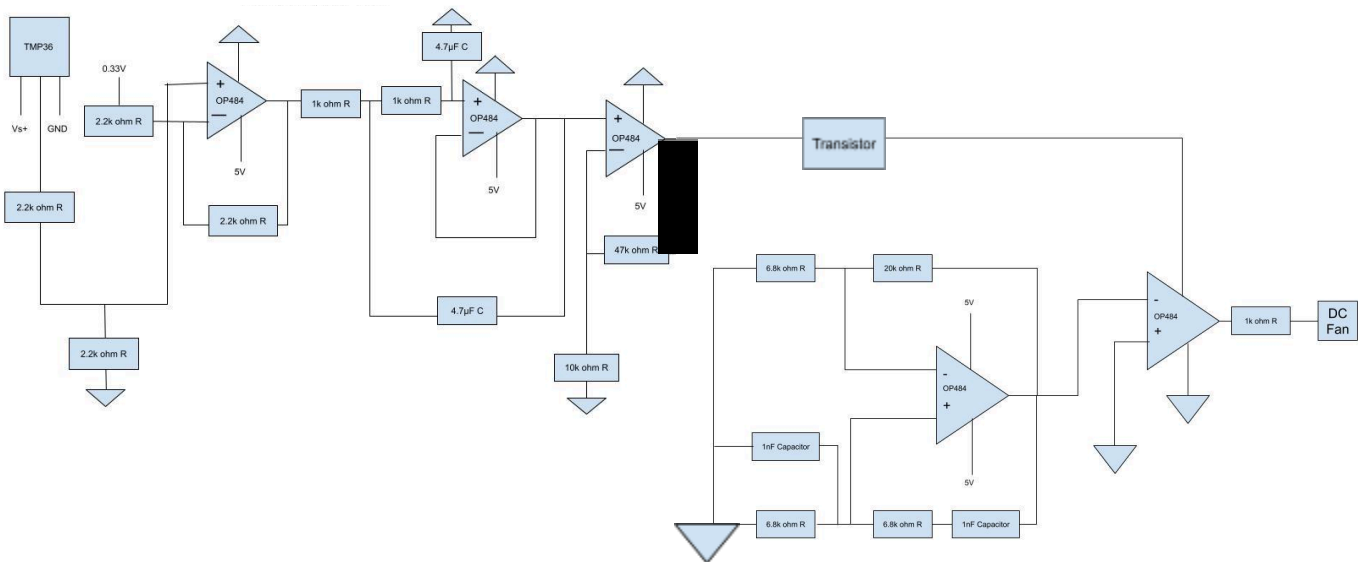
Description

Schematic:



Pictured above is the LTSpice schematic of our circuit. In the schematic, we have our main 5 blocks of our circuit: the differential amplifier, the second-order active filter, the non-inverting amplifier, the Wien Bridge oscillator circuit, and the comparator. Differences between our schematic and our physical circuit include the TMP36 (temperature sensor), which is not a component in LTSpice, along with our 5V DC brushless fan and our transistor. In our simulation, we replaced the temperature sensor with a voltage source that scales at the same rate as our temperature sensor, and then we replaced our fan with an LED. Since we replaced the fan with the LED, we did not need to replace the transistor with anything because the current in our circuit would be sufficient enough to power an LED. The purpose of the transistor in our physical circuit is to scale the current to a sufficient amplitude to power our 5V DC fan.

Block Diagram:



The above diagram shows a high-level block representation of our circuit. It shows that the output voltage of the temperature sensor is compared with a threshold voltage via a differential amplifier. The output of the differential amplifier is then fed into a second-order active filter to reduce high-frequency noise. That filtered voltage is then fed into a non-inverting amplifier, which will amplify our voltage to roughly the 3-5V range, which is sufficient to power our DC fan. The amplified voltage is then passed through a transistor, which amplifies the current to a sufficient amplitude to power our DC fan. Lastly, the output of the transistor is then fed into an oscillator circuit, which makes the motor of the fan turn on and off at smooth rates.

High-level Description of MS3:

In MS3, we made major improvements to our circuit from the previous two milestones. Firstly, we replaced our LED, which represented the change in temperature, with a DC fan. In order for the fan to be powered sufficiently and smoothly, we needed to add a transistor and a pulse-driving circuit.

Before we implemented the transistor into our circuit, the current was in the 20–40mA range. With the addition of the transistor to our circuit, the current is amplified to surpass the 200mA threshold our DC fan needs to operate.

Since we are no longer powering an LED and are instead powering a DC fan, we need a high-frequency Wien Bridge oscillator circuit to help the fan run smoothly. DC motors use electromagnetic coils that have an inherent inductance. On top of this, the fan itself has physical inertia. This causes the fan to stall out when the voltage is low. To combat this, a high-frequency pulse signal provides the “push” it needs. The motor cannot respond to each individual pulse, so

it effectively responds to the average power of the signal, making it behave like a continuous input. For this to work, the frequency should be in the range of 20 to 25 kHz.

We run the oscillator circuit into a comparator to create a square wave that bounces between 0V and the input voltage (the output voltage of the amplifier) at a frequency between 20 to 25kHz.

Lastly, we replaced the first-order passive filter from MS2 with a second-order active filter. The second-order active filter doubles the roll-off rate from the first-order passive filter in MS2 from -20dB/dec to -40dB/dec. On top of this, the original block after the passive filter in MS2 would draw a small amount of current from the passive filter, affecting the filter's operation and shifting the cutoff frequency. The new active filter avoids this issue. We used 1kΩ resistors and 4.7 μF capacitors in this filter, which creates a cutoff frequency of approximately 33.86 Hz using the formula $f_c = 1/(2\pi RC)$.

Description of related pre-existing applications:

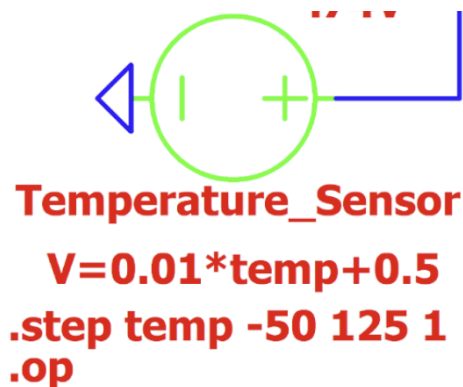
A current existing application would be a HVAC Control Module. Although HVAC focuses on System Protection, Energy Management, and Component Coordination, we are only working on the Temperature Regulation aspect. How this system works is it constantly monitors and adjusts the current environment of the building and heats or cools to keep the constant temperature set by the user. They also use algorithms to prevent temperature spikes to maintain the correct climate [4]. Although our system won't be as complex we still plan on maintaining the basic fundamentals for climate control. We will power our fan based on how high the temperature is to the temperature we have set for our system.

Operation and Design

Input Block: Temperature sensor

Schematic:

The picture below shows the voltage source that represents our temperature sensor in our LTspice simulation. We use the function $V = 0.01 \cdot \text{temp} + 0.5$ to represent the change in output voltage with change in temperature. This function is based on the fact that our temperature sensor scales 10mV for every degree Celsius with an offset of 0.5V. We use a step function with a range of -50–125 degrees Celsius and a step of 1 to represent the proper range that our temperature sensor operates in. The reason we use a voltage source to represent our temperature sensor is because the TMP36 is not a component in the LTspice library.



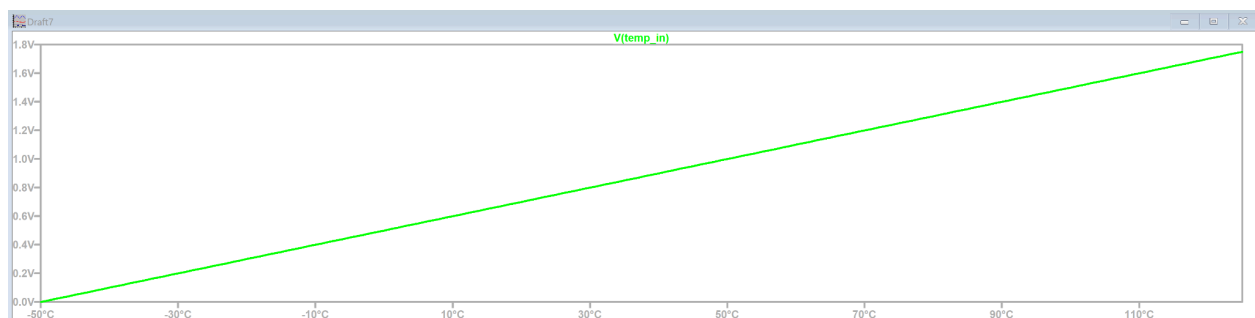
Design Equation:

The TMP36 relates temperature to voltage through the equation:

$$V_{\text{out}} = 0.01 \cdot (\text{degrees Celsius}) + 0.5V [3]$$

The 0.01 is the scaling factor for temperature and there is a 0.5V offset for if the temperature drops below negative. In the schematic above, we simulated this using a voltage source. We inputted the equation and used a .step function to go from -50 to 125 degrees celsius.

Input/Output Plot:

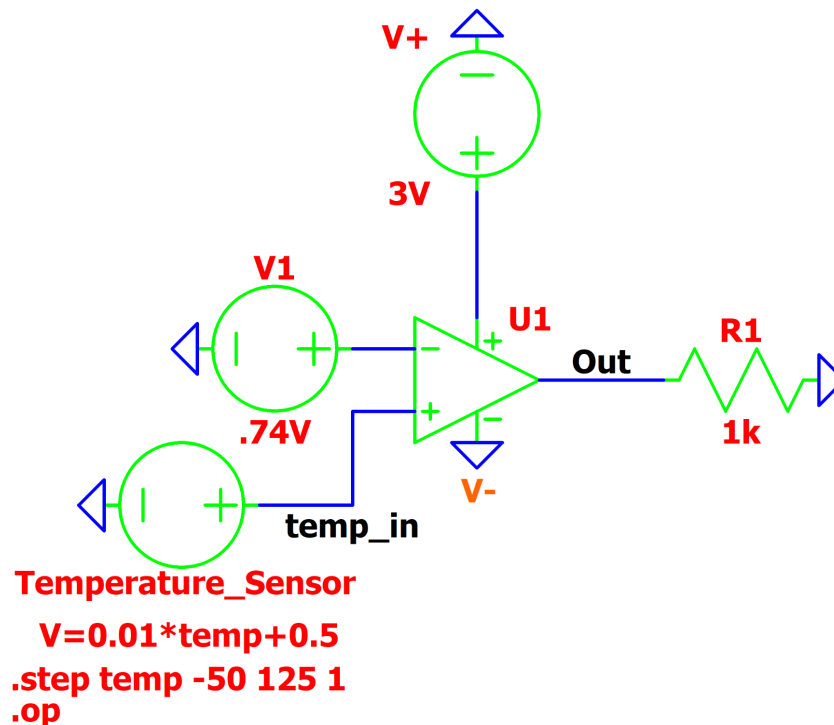


As you can see at 0 degrees celsius, the voltage is 0.5V. This follows the 0.5V offset created. The slope of the line is 0.01(V/C) which is the scalar in the TMP36 V_{out} formula.

MS1 Building Block 1: Op-Amp

Schematic:

Pictured below is the LTspice schematic of our op-amp as a comparator used in MS1. We compare the output voltage of the temperature sensor to predetermined threshold voltage that would indicate a temperature slightly above room temperature. When the output of the temperature sensor is greater than the threshold voltage, 3V is output from the comparator. When the output of the temperature sensor is lower than the threshold voltage, 0V is output from the comparator.



Design Equation:

If $V(\text{Temperature_Sensor}) > V1$: $V_{out} = V+ = 3V$

If $V(\text{Temperature_Sensor}) < V1$: $V_{out} = V- = 0V$

Discussion of Component Choices:

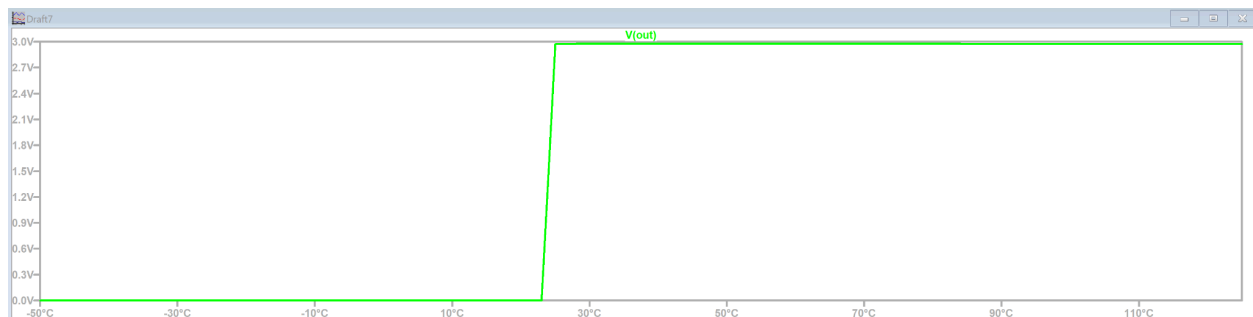
1k ohm Resistor: We made our decision to use the 1k ohm resistor based off the appropriate amount of current needed to dimly light the LED. Typically LEDs operate at 10-20mA [2], however, if the LED drew too much current it would load the op-amp output since the OP97 is a non-rail-to-rail op-amp. We found through taking measurements with our handheld multimeter and making calculations using ohm's law that the best resistor to use would be the 1k ohm resistor since it would have no effect on the output voltage of the op-amp and still dimly light the LED [1].

Op97: We chose to use the Op97 as our comparator because of its simple and safe design. The

Op97 can also operate from a single supply, making it easy to use. And lastly, the Op97 was found in our analog discovery parts kit, making it convenient to use [1].

TMP36: The TMP36 temperature sensor was chosen because its output voltage scales linearly with temperature, making it easy to interface with the comparator circuit. The device has a temperature range of -40°C to $+125^{\circ}\text{C}$, which is more than sufficient for measuring room temperature and slightly higher temperatures. Another advantage of the TMP36 is its built-in 500 mV offset, which ensures that the output voltage remains positive even at negative temperatures. The sensor also provides good accuracy (approximately $\pm 1^{\circ}\text{C}$ typical) and a scale factor of $10\text{ mV}/^{\circ}\text{C}$, making it both reliable and easy to use for temperature measurement in our circuit [3].

Input/Output Plot

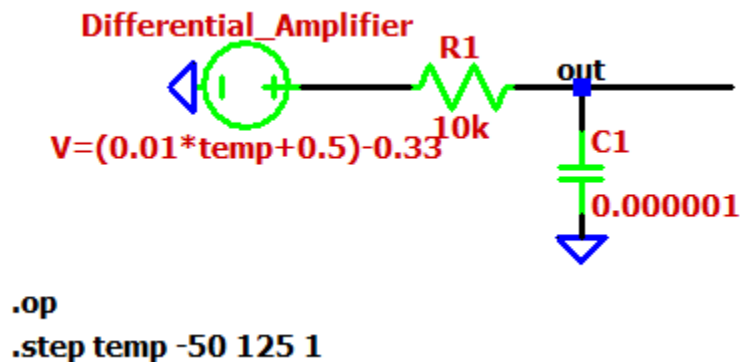


As you can see in the plot, the system works exactly how we intended. We set V_1 to 0.74V, which correlated to 24 degrees Celsius. In the graph, once the temperature reached 24 degrees, the output voltage jumped from 0V to the intended 3V.

MS2 Building Block 1: First Order Passive Filter

Schematic:

Pictured below is the LTspice schematic of our first-order passive filter used in MS2. We replaced the input voltage (the output voltage of our differential amplifier) with a voltage source that scales at the same rate as the output of the differential amplifier. The only difference between the image below and our overall schematic from MS2 is that when we replace the differential amplifier with a voltage source, we cannot properly represent that the output voltage has a max of 5V and a min of 0V. We chose to use a resistor of 10k ohms and a capacitor of 1 μ F because it would provide us with an appropriate cutoff frequency for our circuit of roughly 15.9Hz which we calculated with the function $F_c = 1/(2\pi RC)$.



Design Equation:

$$F_c = 1/(2\pi RC)$$

F_c = Cutoff frequency of a first-order RC low-pass filter

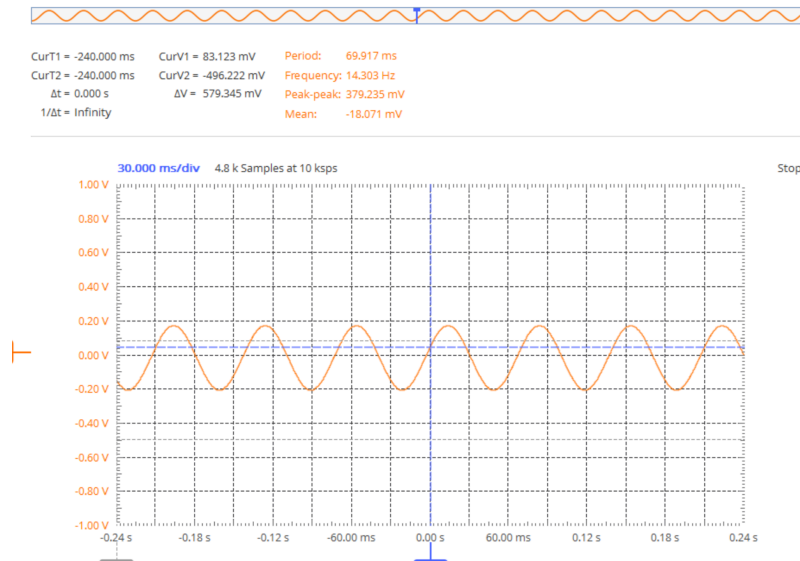
Discussion of Component Choices:

1 μ Capacitor: We chose to use a 1 μ capacitor for our filter since when it is paired with the 10k ohm resistor, it results in a cutoff frequency of 15.9Hz based on the standard cutoff frequency equation. This cutoff frequency allows low-frequency signals to pass while attenuating higher-frequency noise. This is useful for our application because temperature changes occur very slowly (on the order of 0–1 Hz), meaning they pass through the filter with minimal attenuation. At the same time, higher-frequency noise from the OP484 and surrounding circuitry is reduced. We also chose the 1 μ F capacitor because it was readily available in our parts kit.

10k Ohm Resistor: We chose to use a 10k Ω resistor in order to achieve the desired cutoff frequency when combined with the 1 μ F capacitor. Together, these components produce a cutoff frequency of approximately 15.9 Hz, which effectively attenuates high-frequency noise while preserving the temperature signal. Additionally, the 10k Ω resistor was convenient to use since it was included in our parts kit.

Input/Output Plot:

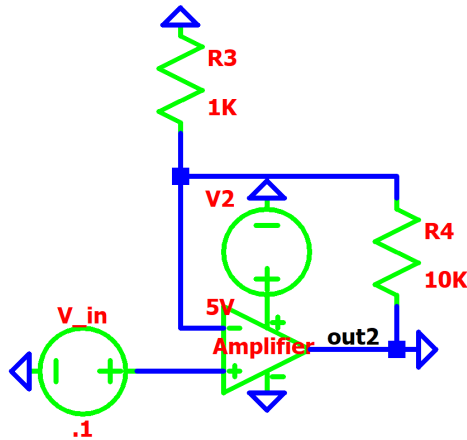
The plot below shows the frequency of 14.303 Hz. The filter is set to filter everything above the threshold of 15.9 Hz. As you can see in the graph below, the noise is filtered and below the threshold.



MS2 Building Block 2: Non-inverting Amplifier

Schematic:

Pictured below is the LTspice schematic of our non-inverting amplifier used in MS2. It takes the output of the first-order passive filter and scales it by a gain of 11, through the use of a 1k ohm and a 10k ohm resistor. The only difference between our schematic below and the schematic of the overall circuit for MS2 is that we had to replace the input voltage of the amplifier (output voltage of the filter) with a voltage source. We chose a value of 0.1V because it would be easy to use experimentally to prove that our non-inverting amplifier worked as intended.



Design Equation:

$$V_{out2} = (1 + R4/R3) * V_{in+}$$

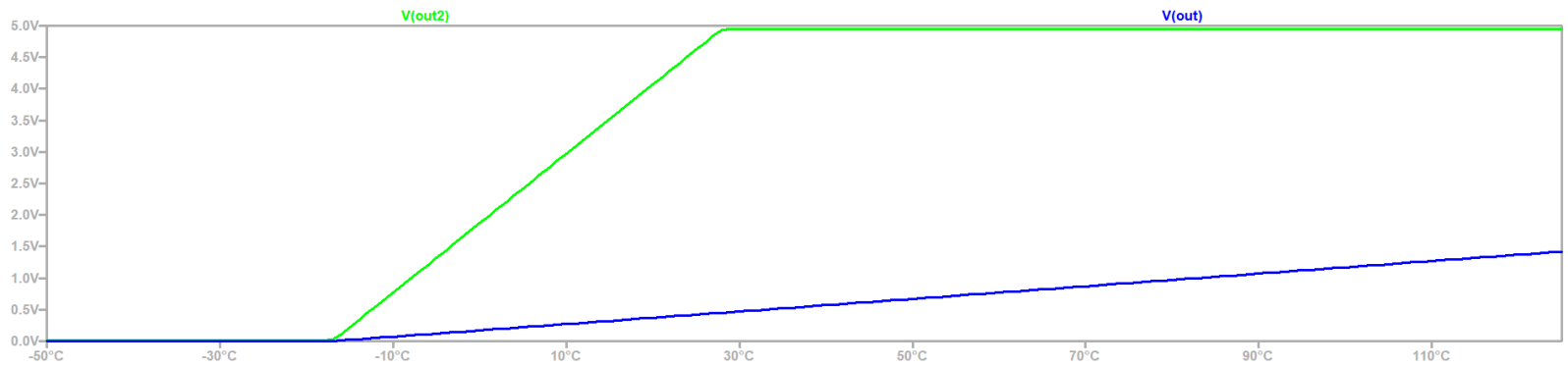
Discussion of Component Choices:

OP97: We chose to use the OP97 for our non-inverting amplifier due to its simplicity and availability in our parts kit. We had previously used the OP97 in MS1 as a comparator, so we were already familiar with its pin configuration, operation, and limitations, including the fact that it is not rail-to-rail. In this configuration, the output voltage from the first-order passive filter is fed into the non-inverting input (V_{in+}). The op-amp is powered with +5V and ground, which sets the upper and lower bounds of the output voltage. However, due to the OP97 not being rail-to-rail, the output cannot reach the supply rails and will saturate slightly below 5V. We used two resistors (R₃ = 1kΩ and R₄ = 10kΩ) to configure the circuit as a non-inverting amplifier based on the equation for V_{out2}, which results in a gain of 11 [1].

Resistors: For our non-inverting amplifier, we selected resistor values of 1k ohms for R₃ and 10k ohms for R₄ to achieve a gain of 11x. The input voltage from the previous stage is approximately 0–0.5V for realistic temperature ranges. After amplification, this produces an output of approximately 0–5V (slightly less in practice due to op-amp limitations). This output range is sufficient to drive the LED, which begins to turn on at approximately 1.7V [2].

Input/Output Plot:

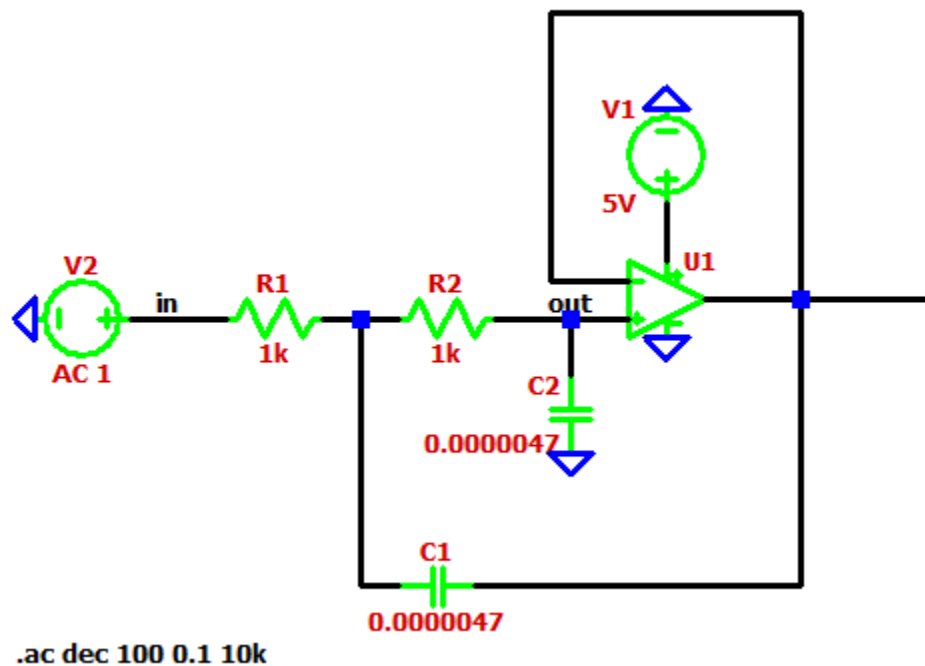
The image below shows the output graph from LTspice. The V(out) trace represents the output voltage from the differential amplifier of our circuit, or the input voltage to the non-inverting amplifier. The V(out2) trace represents the output voltage from the non-inverting amplifier. As you can see from the graph, the non-inverting amplifier scales the input voltage by a gain of 11, with a min of 0V and a max of 5V.



MS3 Building Block 1: Second-Order Active Filter

Schematic:

Pictured below is the LTspice schematic of our Second-order active filter. We have two resistors of equal resistance in series connected to the Vin+ of the OP484, and then two capacitors of equal capacitance. The first capacitor is connected between the two resistors and to the output of the OP484. The second resistor is connected to the output of the two resistors and is connected to ground. Also, the Vin- terminal is connected to the output terminal of the OP484 to create a feedback loop.



Design Equation:

Since $R1 = R2$, and $C1 = C2$, we get the function below for cutoff frequency

$$F_c = 1/(2\pi RC)$$

Discussion of Component Choices:

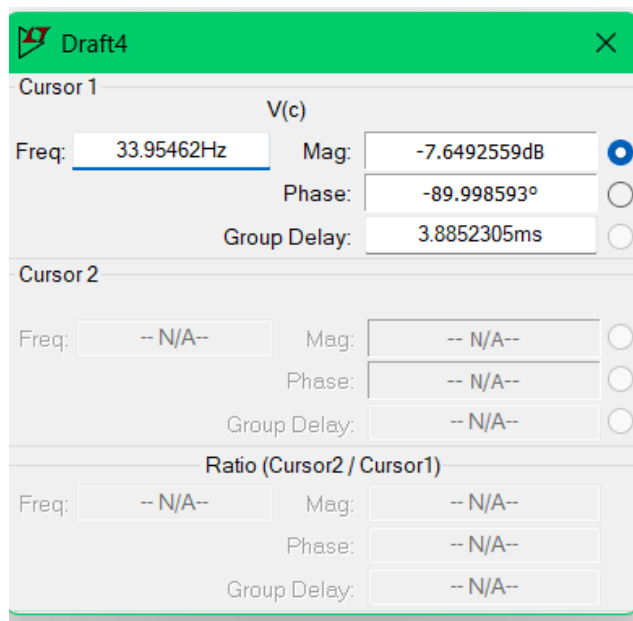
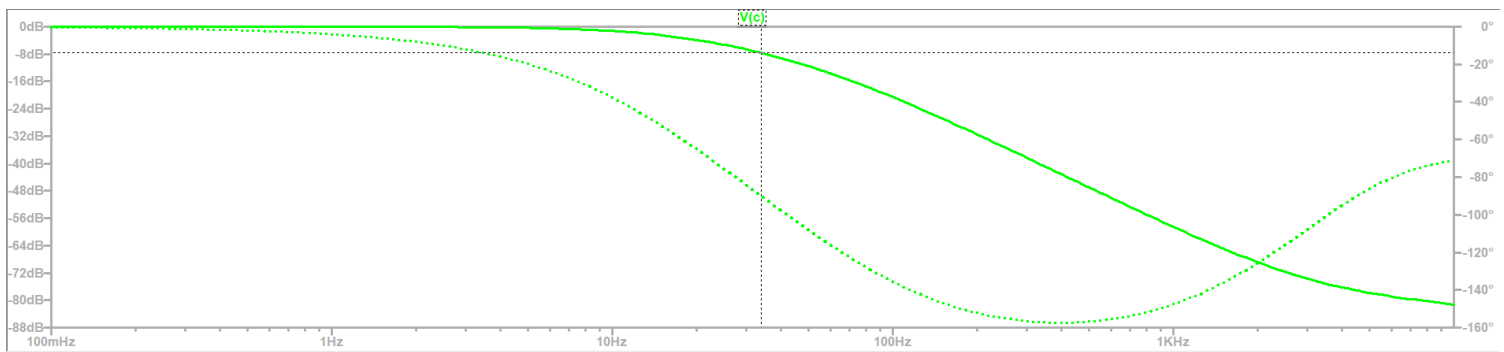
Resistors & Capacitors: For our second-order active filter, we chose to use resistors of equal resistance (1k ohm) and capacitors of equal capacitance (4.7μF). We chose to make all of the components relatively equal because it would result in the filter having a gain of one. If we had chosen to use resistors or capacitors of differing values, the circuit gain would be affected. This is important because when the gain of second-order active filters becomes greater than 1, the circuit becomes unstable, and when it becomes greater than 3 it turns into an oscillator circuit. With the values of 1k ohm resistors and 4.7μF capacitors, our cutoff frequency becomes roughly 34Hz, which is an appropriate cutoff frequency given our circuit.

OP484: For our second-order active filter we chose to use the OP484 because it is a rail-to-rail op-amp. Since we are working with the pre-amplified voltage, we are working at very low

voltages (less than 1V). When our voltages are this low, a non-rail-to-rail op-amp won't be able to properly recognize these low voltages. That is why we decided to use the OP484 as the op-amp for our second-order active filter. Another reason we chose to use the OP484 specifically is because of its convenience already being in our parts kit [5].

Input/Output Plot:

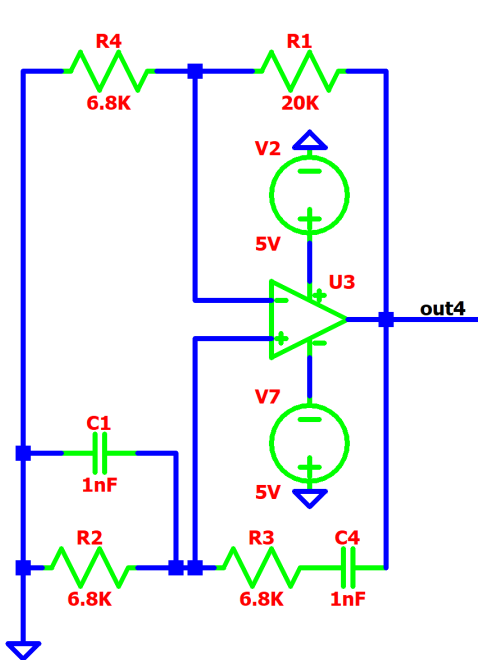
Pictured below is the resulting Bode plot of our second-order active filter. The bode plot has a flat gain at low frequencies, a cutoff around ~30–40 Hz, and a roll-off of approximately -40 dB/decade with phase shifting toward -180°.



MS3 Building Block 2: Oscillator Circuit

Schematic:

Pictured below is the LTspice schematic of our Wien Bridge oscillator circuit used in MS3. The oscillator generates a high-frequency sinusoidal waveform that operates at approximately 23.4 kHz. We chose equal resistor values of 6.8kΩ and equal capacitor values of 1nF in the Wien bridge feedback network in order to set the oscillation frequency within the desired 20–25 kHz range for smooth operation of the DC fan motor. The oscillator also uses a non-inverting amplifier configuration with a gain greater than 3 to satisfy the oscillation condition required for sustained output. The output sinusoidal signal is then fed into a comparator stage, which converts the waveform into a square wave used to drive the transistor and fan stage.



Design Equation:

$$f = 1/(2\pi R_3 C_4) = 23.4\text{kHz}$$

$$\text{Gain} = 1 + 20\text{k}/6.8\text{k} = 3.947$$

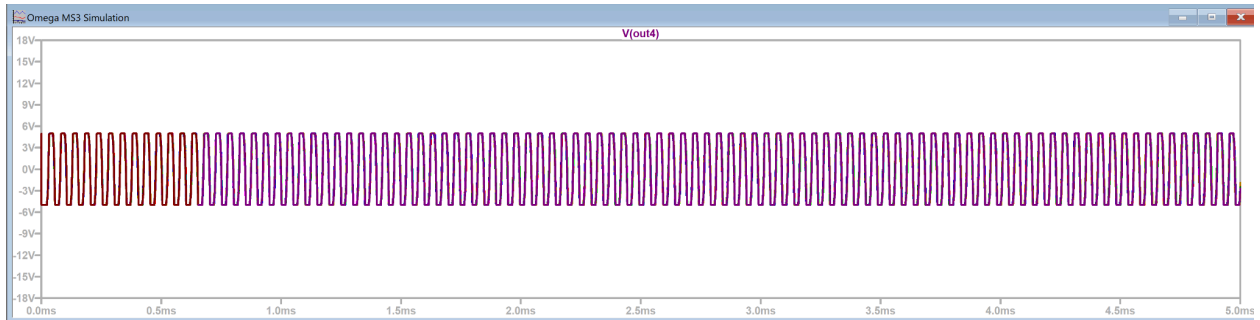
Discussion of Component Choices:

R2=R3=6.8k and C1=C4=1nF. Using the formula $f = 1/(2\pi C R)$, we want R2 and R3 equal and C1 and C4 equal. On top of this, choose the numbers 1nF and 6.8K Ohms to generate a frequency of 23.4kHz which is in the needed range for optimal driving of the DC motor.

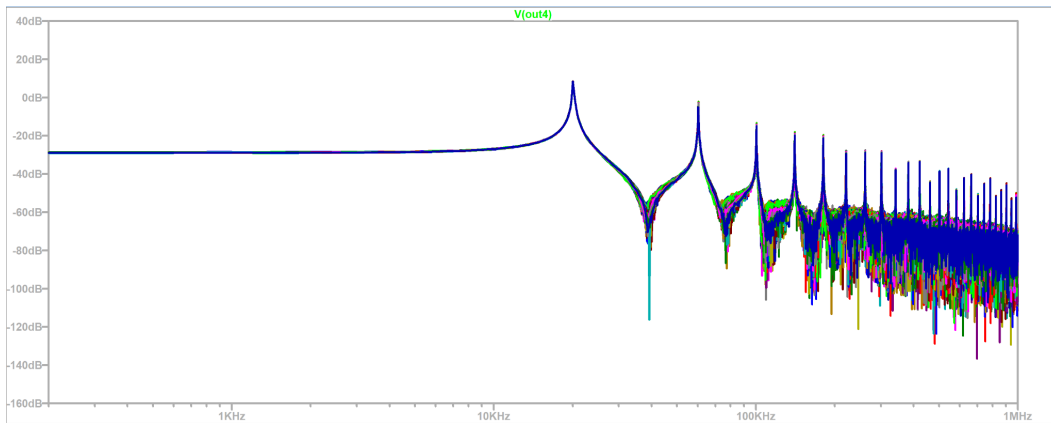
R4=6.8k and R1 = 20k. Using the formula for gain, $1 + R1/R4$, we chose the resistances to have a gain of over 3. This is because the calculated attenuation is $1/3$. Because of this, we want a gain of 3 or above to cancel it out. In this case we have a gain of slightly above 3. This is okay because the gain will increase until it hits the limits of the supply voltages. Because of this it will go on forever at a capped amplitude. If the gain is less than 3, it will converge to 0.

Input/Output Plot:

Pictured below is the output from the oscillator circuit. There is a sine wave with a peak to peak of 10V centered around 0V.



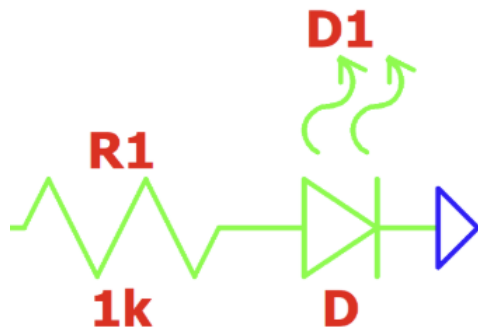
As you can see in the FTT Graph, the highest point is the frequency. This happens around 23kHz which is what was expected.



Output Block: 5V DC Brushless Fan

Schematic:

Pictured below is the LTspice schematic of the LED we use to represent the change in voltage with a change in temperature in our simulation. We use the LED in place of the fan since there is no 5V fan in the LTspice component library. Our fan operates in the range of roughly 3-5V and around 200mA or greater [6].



Design Equation:

gain(non-inverting amplifier) = 5.7

$V_{out}(\text{differential amplifier}) = (0.01V \cdot (\text{Temperature in Celsius}) + 0.5V) - 0.33V$

$V_{out}(\text{non-inverting amplifier}) = V_{out}(\text{differential_amplifier}) \cdot \text{gain}(\text{non-inverting amplifier})$

$f(\text{oscillator}) = 1/(2\pi R \cdot C) \approx 23.4\text{kHz}$

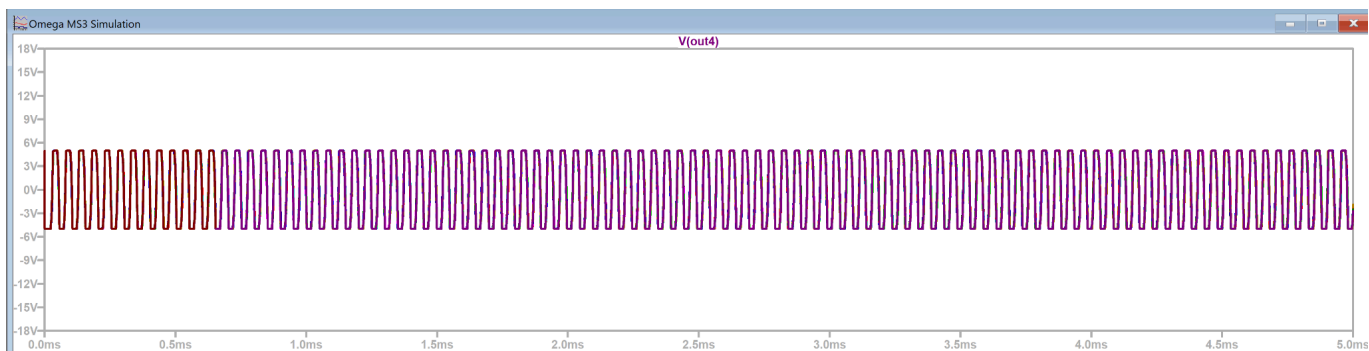
Transistor gain = $hFE = 25$ to 50 [7]

$I_b = 20\text{--}40\text{mA}$

$I_{out}(\text{transistor}) = hFE \cdot I_b$ [7]

Input/Output Plot:

Pictured below is the LTspice plot of change in voltage compared to change in time after the oscillator circuit. This is the voltage that is going to be fed into our 5V DC fan.



Integration and Optimization

There are 6 main stages to our circuit for MS3: sensing (input), comparing (differential amplifier), filtering (MS3 block 1), amplifying (MS2 block 2), pulse-driving (MS3 block 2), and indicating (output).

The sensing phase consists of the TMP36, which is supplied 5V, with a voltage output based on the surrounding temperature. We chose this sensor because it outputs a scaled voltage, which is very easy to work with given our goal. The output voltage of the sensor is then fed into Vin+ (pin 3) of our OP484, which we utilize as a differential amplifier during our comparing phase. In MS1, a comparator was used, which caused the output to switch abruptly between two voltage levels when a threshold was crossed. For MS2 and MS3, we instead use a differential amplifier so that the output voltage is proportional to the difference between the sensor voltage and a reference voltage, allowing for a smooth and continuous response to temperature changes. The OP484 was chosen because it is a rail-to-rail op-amp, allowing for low voltages to be outputted, which is important for maximizing the usable output range in our circuit [5].

The V+ terminal (pin 4) of the OP484 is supplied with 5V, and the V- terminal (pin 11) is connected to ground. A predetermined reference voltage is supplied to Vin- (pin 2) [5]. We used four equal resistors to configure the op-amp as a differential amplifier. Because all resistors are equal, the circuit has a gain of 1, meaning the output directly represents the difference between the input signal and the reference voltage. We intentionally avoided amplifying at this stage so that the noise introduced by the op-amp would not be amplified before filtering.

Our next stage consists of two 1kΩ resistors and two 4.7μF capacitors. The resistors are placed in series with the output of the differential amplifier and connected to the Vin+ terminal of the OP484. One capacitor is connected from the output node of the resistors in series to ground. The other is connected between the two resistors in series to the output of the op-amp (OP484). And lastly, the Vin- terminal is connected to the output of the OP484 to create a feedback loop. Together, these components form a second-order active filter. This filter attenuates high-frequency noise generated by the op-amp while allowing the low-frequency temperature signal to pass. The reason we replaced our first-order passive filter in MS2 with a second-order active filter in MS3 is because the original block after the passive filter in MS2 would draw a small amount of current from the passive filter, affecting the filter's operation and shifting the cutoff frequency. The new active filter avoids this issue.

The next stage is the amplifying stage, which utilizes an OP484 configured as a non-inverting amplifier. The filtered signal is fed into the Vin+ terminal of the op-amp, and the op-amp is powered with +5V and ground, setting the output range approximately between 0V and 5V. We use a 10kΩ resistor and a 47kΩ resistor to set the gain to be roughly 5.7, based on the equation $V_{out2} = (1 + R_4/R_3)V_{in+}$. This amplification increases the signal to a level sufficient to drive the motor.

Our next stage consists of our transistor and pulse-driving circuit. Since we are no longer powering an LED and are instead powering a DC motor fan, we need a greater current to be able to properly power the fan. To do this, we use the TIP31CFS NPN to increase the current in our circuit from the 20–40mA range to surpass the 200mA threshold our fan needs to run [6].

A high-frequency pulse signal (approximately 20–25 kHz) is applied to this stage using a Wien Bridge oscillator circuit. This creates a sine wave that is fed into the IN- of a comparator. Since the sine wave is centered around 0V, and IN+ is attached to ground, the comparator creates a square wave switching between V+ and V-. V+ is connected to the output of the amplifier, and

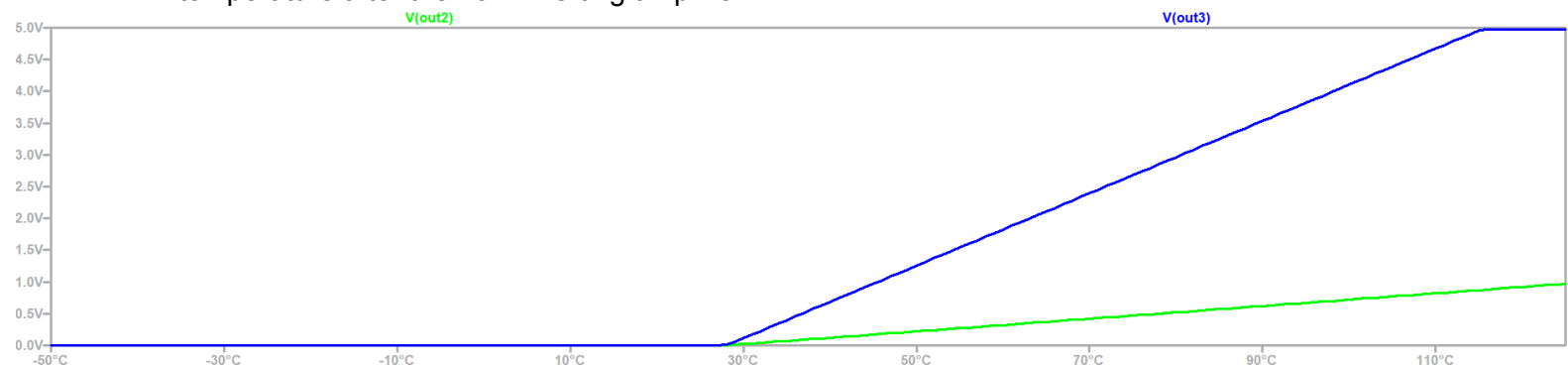
V- is attached to ground. This makes the signal switch between 0V and the output voltage of the non-inverting amplifier, creating a square wave that drives the motor. DC motors use electromagnetic coils that have an inherent inductance, and the fan itself has physical inertia. Because of this, the motor does not respond to each individual pulse. Instead, it responds to the average power of the signal, which results in smooth and continuous motion. This also helps prevent the motor from stalling at lower effective drive levels.

Our last stage is our indicating stage. This phase consists of our 5V DC brushless fan. The high-frequency oscillating output signal is provided to our fan, allowing it to respond smoothly to the control signal despite the rapidly switching input. We chose this specific fan because it operates in the proper voltage range given our voltage source. Our fan operates in roughly the 3–5V range and requires approximately 200mA or greater to run properly [6].

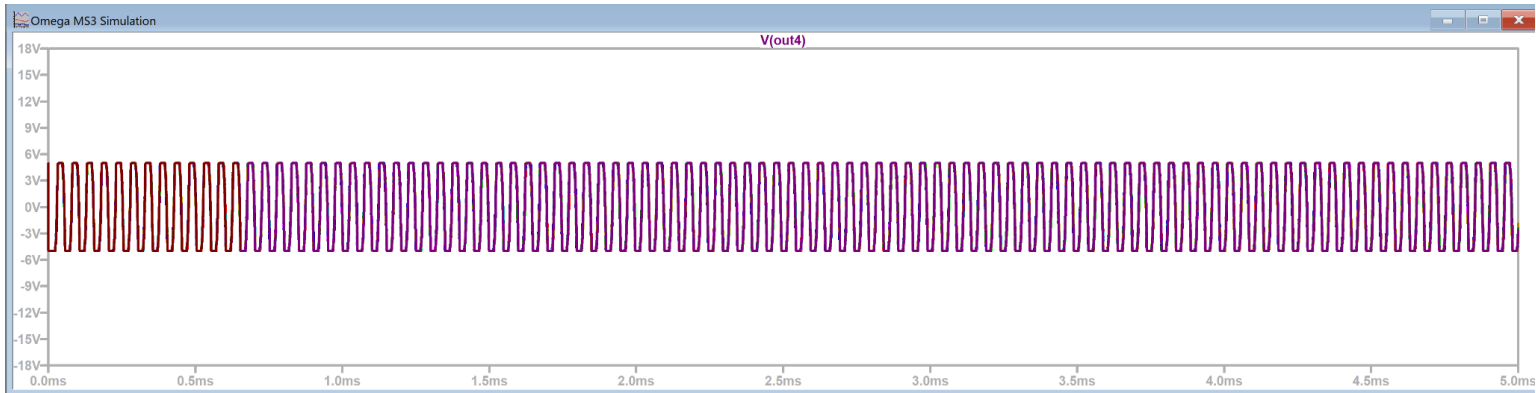
Each stage was designed to ensure compatibility between blocks, with the output of one stage falling within the acceptable input range of the next. This allows the overall system to operate smoothly and reliably.

Input/Output Plot:

Pictured below is the input output plot of our overall circuit, you can see the output voltage compared to change in temperature at three points in our circuit. The V(out2) trace represented by the green line is the output voltage compared to temperature after the second-order active filter. The V(out3) trace represented by the blue line is the output voltage compared to temperature after the non-inverting amplifier.



Pictured below is the square wave that is generated by our oscillator circuit and that is the input voltage of our fan.



Operating Conditions

Limitations:

One limitation of our circuit is the use of a fixed duty cycle in our pulse-driving stage. Since the duty cycle is constant (50%), the motor does not fully scale its speed with changes in temperature. This limits the system's ability to provide precise control over the fan speed, especially at lower or higher temperature ranges.

Another limitation is the delay introduced by the second-order active filter. While the filter effectively removes high-frequency noise, it also slows the system's response to rapid temperature changes. As a result, the output signal does not instantly reflect sudden changes in temperature, which can affect how quickly the motor responds.

Another limitation is the size and power capability of the fan being used. Our fan is relatively small (approximately 30mm × 30mm), which limits the amount of airflow it can produce. While it is sufficient for demonstrating the functionality of our system, it may not be effective for larger-scale cooling applications or situations requiring significant airflow.

Tradeoff:

A key tradeoff in our design is between smooth motor operation and accurate signal representation. By using a high-frequency pulse signal, we ensure that the motor runs smoothly and does not stall at lower effective drive levels. However, this comes at the cost of losing a direct analog relationship between voltage and motor speed, since the output is now a pulsed signal rather than a purely continuous one.

Possible Improvements:

One possible improvement would be to implement pulse-width modulation (PWM) control, where the duty cycle of the pulse signal is adjusted based on the temperature. This would allow the motor speed to scale more accurately with temperature, improving overall system performance.

Another improvement would be to replace the transistor with a more efficient switching device, such as a MOSFET. This would reduce power loss and improve efficiency when driving the motor, especially at higher current levels.

A further improvement would be to optimize the cutoff frequency of the active filter to better balance noise reduction and response time, particularly when accounting for noise introduced by the motor and switching circuitry.

ABET 3.2 Engineering Design Considerations

In designing this circuit, we selected components and parameters to ensure safe and efficient operation while meeting the goal of temperature-based control. The system operates at a low voltage of 5V, which reduces safety risks and power consumption, making it suitable for consumer and small-scale applications. The use of a transistor and pulse-driving circuit allows

the motor to operate efficiently without excessive power loss, improving reliability and minimizing heat generation. From an economic and environmental perspective, we used readily available components from our parts kit to reduce cost and material waste, and the use of a small 30mm × 30mm fan helps limit overall power consumption. While our design is relatively simple, it demonstrates consideration of safety, efficiency, cost, and environmental impact, which are important factors in real-world engineering applications.

References

[1] Analog Devices, "OP97: Low Power, High Precision Operational Amplifier," Datasheet.

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